Web Application Packet Analysis

Problem:

The payroll website used by employees for filing their time-in and time-out records is experiencing excessive loading times before reaching the login page. In addition, navigating between pages within the website is significantly delayed. For example, moving from the Time-In page to the Time-In Records page typically takes approximately with the average of 3 to 5 minutes.

Factors that were checked before packet analysis:

Internet Speed

A. The internet connection was tested and found to be operating within normal parameters. The download speed was recorded at 377 Mbps, while the upload speed was recorded at 293 Mbps. These values are generally sufficient for normal web browsing and online activities.

B. Is the Issue Recurring?

Yes. The issue was initially reported approximately six months ago and remains unresolved at the time of this investigation.

C. Who Are the Affected Users?

The issue affects all users, including personnel from other departments.

D. Are There Any Announced Updates or Changes That May Have Caused the Problem?

At the time the concern was reported, no updates, maintenance notices, or resolutions had been announced by the website provider.

Observation Table:

The packet analysis collected xxx packets. During the process, notable packets were observed and partially analyzed to create a pattern of packet behaviors during the process of webpage navigation.

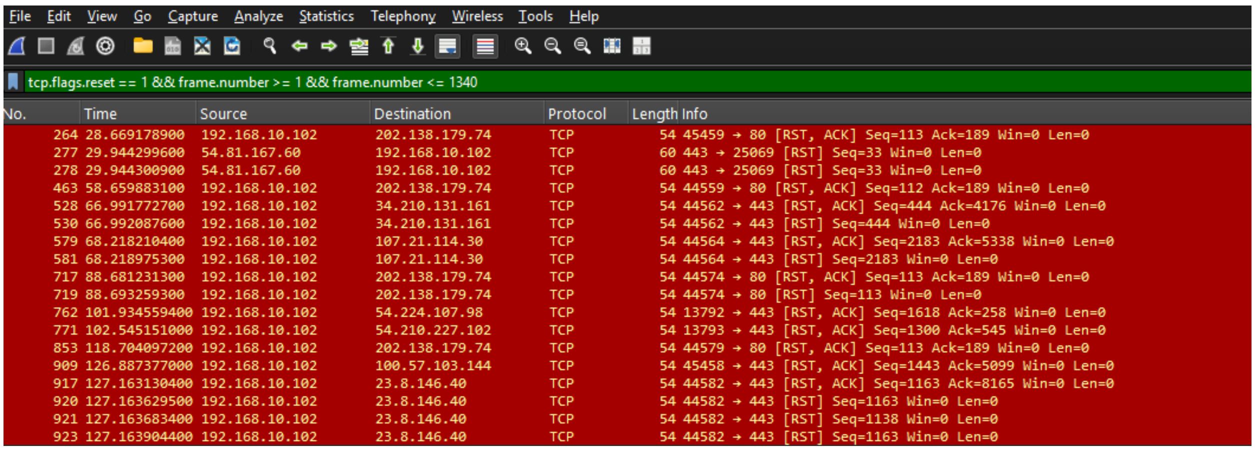
Stage	Application State	Approximate Packet Range	Time Spent
Initial Request	URL entered → Login Portal displayed	264 – 1340	11m 04s
Authentication and Home Page Load	Login Portal → Home Page	1341 – 1398	2m 47s
Navigation to Filed Work List	Home Page → Filed Work List	1399 – 1622	23s
Navigation to Time-In Records	Filed Work List → Time-In Records	1623 – 2512	27s
Returning to Home Page Navigation	Time-In Records → Home Page	2513 – 3834	6m 35s
Total Observed Session	Entire workflow	264 – 3834	21m 16s

Observation in each stage

1. **Initial Request:**

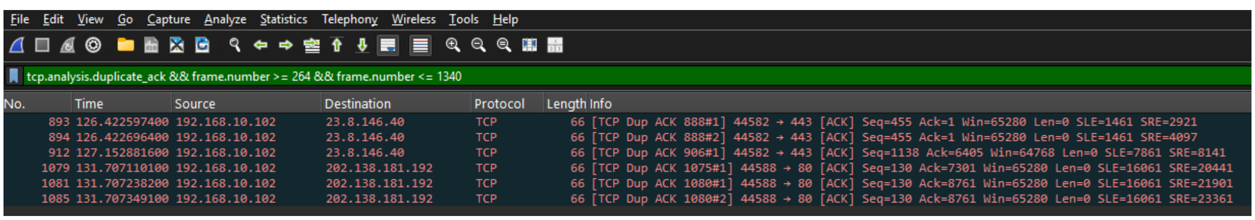
To initiate the process, the website URL was entered into the browser's address bar, and the elapsed time was observed until the Login Portal appeared. For approximately 11 minutes and 4 seconds, the webpage displayed only a loading indicator before the login page became accessible. The login credentials were already saved and automatically recognized by the web application.

This stage covered an estimated packet range of 264–1340, totaling 1,076 captured packets. Within this range, several notable packets were observed:



No.	Time	Source	Destination	Protocol	Length	Info
264	28.669178900	192.168.10.102	202.138.179.74	TCP	54	45459 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
277	29.944299600	54.81.167.60	192.168.10.102	TCP	60	443 → 25069 [RST] Seq=33 Win=0 Len=0
278	29.944300900	54.81.167.60	192.168.10.102	TCP	60	443 → 25069 [RST] Seq=33 Win=0 Len=0
463	58.659883100	192.168.10.102	202.138.179.74	TCP	54	44559 → 80 [RST, ACK] Seq=112 Ack=189 Win=0 Len=0
528	66.991772700	192.168.10.102	34.210.131.161	TCP	54	44562 → 443 [RST, ACK] Seq=444 Ack=4176 Win=0 Len=0
530	66.992087600	192.168.10.102	34.210.131.161	TCP	54	44562 → 443 [RST] Seq=444 Win=0 Len=0
579	68.218210400	192.168.10.102	107.21.114.30	TCP	54	44564 → 443 [RST, ACK] Seq=2183 Ack=5338 Win=0 Len=0
581	68.218975300	192.168.10.102	107.21.114.30	TCP	54	44564 → 443 [RST] Seq=2183 Win=0 Len=0
717	88.681231300	192.168.10.102	202.138.179.74	TCP	54	44574 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
719	88.693259300	192.168.10.102	202.138.179.74	TCP	54	44574 → 80 [RST] Seq=113 Win=0 Len=0
762	101.934559400	192.168.10.102	54.224.107.98	TCP	54	13792 → 443 [RST, ACK] Seq=1618 Ack=258 Win=0 Len=0
771	102.545151000	192.168.10.102	54.210.227.102	TCP	54	13793 → 443 [RST, ACK] Seq=1300 Ack=545 Win=0 Len=0
853	118.704097200	192.168.10.102	202.138.179.74	TCP	54	44579 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
909	126.887377000	192.168.10.102	100.57.103.144	TCP	54	45458 → 443 [RST, ACK] Seq=1443 Ack=5099 Win=0 Len=0
917	127.163130400	192.168.10.102	23.8.146.40	TCP	54	44582 → 443 [RST, ACK] Seq=1163 Ack=8165 Win=0 Len=0
920	127.163629500	192.168.10.102	23.8.146.40	TCP	54	44582 → 443 [RST] Seq=1163 Win=0 Len=0
921	127.163683400	192.168.10.102	23.8.146.40	TCP	54	44582 → 443 [RST] Seq=1138 Win=0 Len=0
923	127.163904400	192.168.10.102	23.8.146.40	TCP	54	44582 → 443 [RST] Seq=1163 Win=0 Len=0

TCP Reset (RST) packets were recorded 18 times between packets 264 and 923.



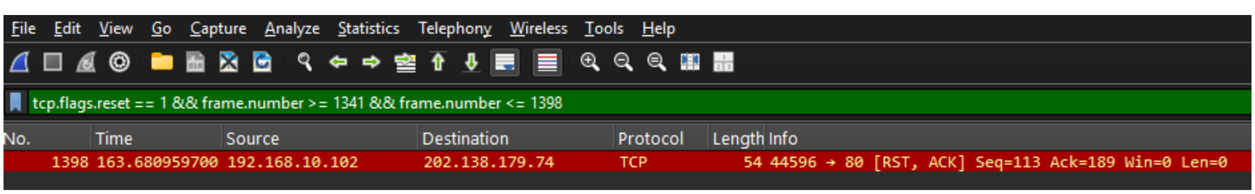
No.	Time	Source	Destination	Protocol	Length	Info
893	126.422597400	192.168.10.102	23.8.146.40	TCP	66	[TCP Dup ACK 888#1] 44582 → 443 [ACK] Seq=455 Ack=1 Win=65280 Len=0 SLE=1461 SRE=2921
894	126.422696400	192.168.10.102	23.8.146.40	TCP	66	[TCP Dup ACK 888#2] 44582 → 443 [ACK] Seq=455 Ack=1 Win=65280 Len=0 SLE=1461 SRE=4097
912	127.152881600	192.168.10.102	23.8.146.40	TCP	66	[TCP Dup ACK 906#1] 44582 → 443 [ACK] Seq=1138 Ack=6405 Win=64768 Len=0 SLE=7861 SRE=8141
1079	131.707110100	192.168.10.102	202.138.181.192	TCP	66	[TCP Dup ACK 1075#1] 44588 → 80 [ACK] Seq=130 Ack=7301 Win=65280 Len=0 SLE=16061 SRE=20441
1081	131.707238200	192.168.10.102	202.138.181.192	TCP	66	[TCP Dup ACK 1080#1] 44588 → 80 [ACK] Seq=130 Ack=8761 Win=65280 Len=0 SLE=16061 SRE=21901
1085	131.707349100	192.168.10.102	202.138.181.192	TCP	66	[TCP Dup ACK 1080#2] 44588 → 80 [ACK] Seq=130 Ack=8761 Win=65280 Len=0 SLE=16061 SRE=23361

TCP Duplicate Acknowledgement (Dup ACK) packets were observed 6 times between packets 893 and 1085.

2. **Authentication and Home Page Load:**

After the Login Portal appeared, the web application was accessed using the saved credentials. The page remained in a loading state for approximately 2 minutes and 47 seconds before the Home Page was displayed. This stage covered an approximate packet range of 1341–1398, totaling 58 captured packets.

Notable packet observations:



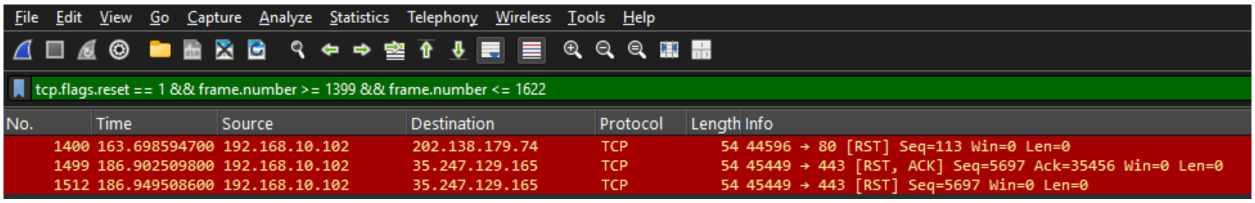
No.	Time	Source	Destination	Protocol	Length	Info
1398	163.680959700	192.168.10.102	202.138.179.74	TCP	54	44596 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0

One TCP Reset (RST) packet was observed in packet 1398..

3. Navigation to Filed Work List:

Navigation from the Home Page to the Filed Work List page took approximately 23 seconds. This webpage transition covered the packet range of 1399–1622, totaling 224 captured packets.

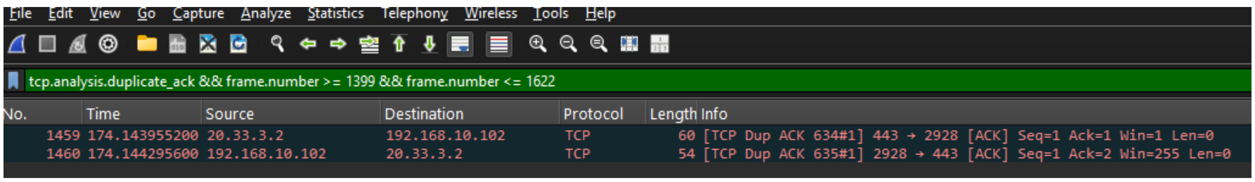
Notable packet observations:



tcp.flags.reset == 1 && frame.number >= 1399 && frame.number <= 1622

No.	Time	Source	Destination	Protocol	Length	Info
1400	163.698594700	192.168.10.102	202.138.179.74	TCP	54	44596 → 80 [RST] Seq=113 Win=0 Len=0
1499	186.902509800	192.168.10.102	35.247.129.165	TCP	54	45449 → 443 [RST, ACK] Seq=5697 Ack=35456 Win=0 Len=0
1512	186.949508600	192.168.10.102	35.247.129.165	TCP	54	45449 → 443 [RST] Seq=5697 Win=0 Len=0

TCP Reset (RST) packets were observed 3 times between packets 1400 and 1512.



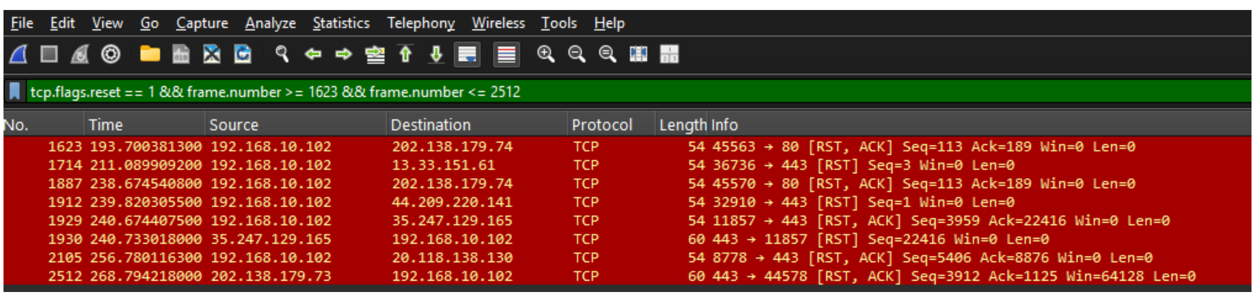
tcp.analysis.duplicate_ack && frame.number >= 1399 && frame.number <= 1622

No.	Time	Source	Destination	Protocol	Length	Info
1459	174.143955200	20.33.3.2	192.168.10.102	TCP	60	[TCP Dup ACK 634#1] 443 → 2928 [ACK] Seq=1 Ack=1 Win=1 Len=0
1460	174.144295600	192.168.10.102	20.33.3.2	TCP	54	[TCP Dup ACK 635#1] 2928 → 443 [ACK] Seq=1 Ack=2 Win=255 Len=0

TCP Duplicate Acknowledgement (Dup ACK) packets were observed twice in packets 1459 and 1460.

4. Navigation to Time-In Records:

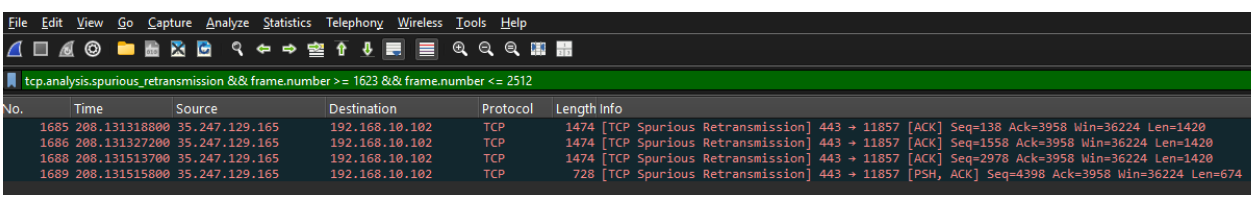
Navigation from the Filed Work List page to the Time-In Records page took approximately 27 seconds. This stage covered the packet range of 1623–2512, totaling 890 captured packets.
Notable packet observations:



tcp.flags.reset == 1 && frame.number >= 1623 && frame.number <= 2512

No.	Time	Source	Destination	Protocol	Length	Info
1623	193.700381300	192.168.10.102	202.138.179.74	TCP	54	45563 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
1714	211.089909200	192.168.10.102	13.33.151.61	TCP	54	36736 → 443 [RST] Seq=3 Win=0 Len=0
1887	238.674540800	192.168.10.102	202.138.179.74	TCP	54	45570 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
1912	239.820305500	192.168.10.102	44.209.220.141	TCP	54	32910 → 443 [RST] Seq=1 Win=0 Len=0
1929	240.674407500	192.168.10.102	35.247.129.165	TCP	54	11857 → 443 [RST, ACK] Seq=3959 Ack=22416 Win=0 Len=0
1930	240.733018000	35.247.129.165	192.168.10.102	TCP	60	443 → 11857 [RST] Seq=22416 Win=0 Len=0
2105	256.780116300	192.168.10.102	20.118.138.130	TCP	54	8778 → 443 [RST, ACK] Seq=5406 Ack=8876 Win=0 Len=0
2512	268.794218000	202.138.179.73	192.168.10.102	TCP	60	443 → 44578 [RST, ACK] Seq=3912 Ack=1125 Win=64128 Len=0

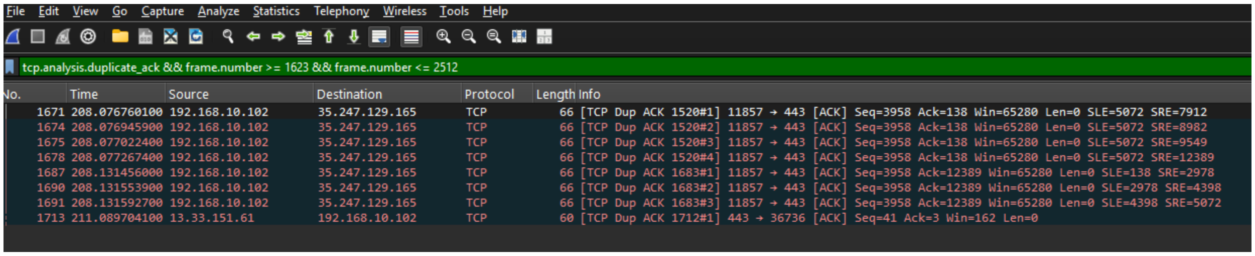
TCP Reset (RST) packets were observed 8 times between packets 1623 and 2512.



tcp.analysis.spurious_retransmission && frame.number >= 1623 && frame.number <= 2512

No.	Time	Source	Destination	Protocol	Length	Info
1685	208.131318800	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 11857 [ACK] Seq=138 Ack=3958 Win=36224 Len=1420
1686	208.131327200	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 11857 [ACK] Seq=1558 Ack=3958 Win=36224 Len=1420
1688	208.131513700	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 11857 [ACK] Seq=2978 Ack=3958 Win=36224 Len=1420
1689	208.131515800	35.247.129.165	192.168.10.102	TCP	728	[TCP Spurious Retransmission] 443 → 11857 [PSH, ACK] Seq=4398 Ack=3958 Win=36224 Len=674

Spurious Retransmission packets were observed 4 times between packets 1685 and 1689.



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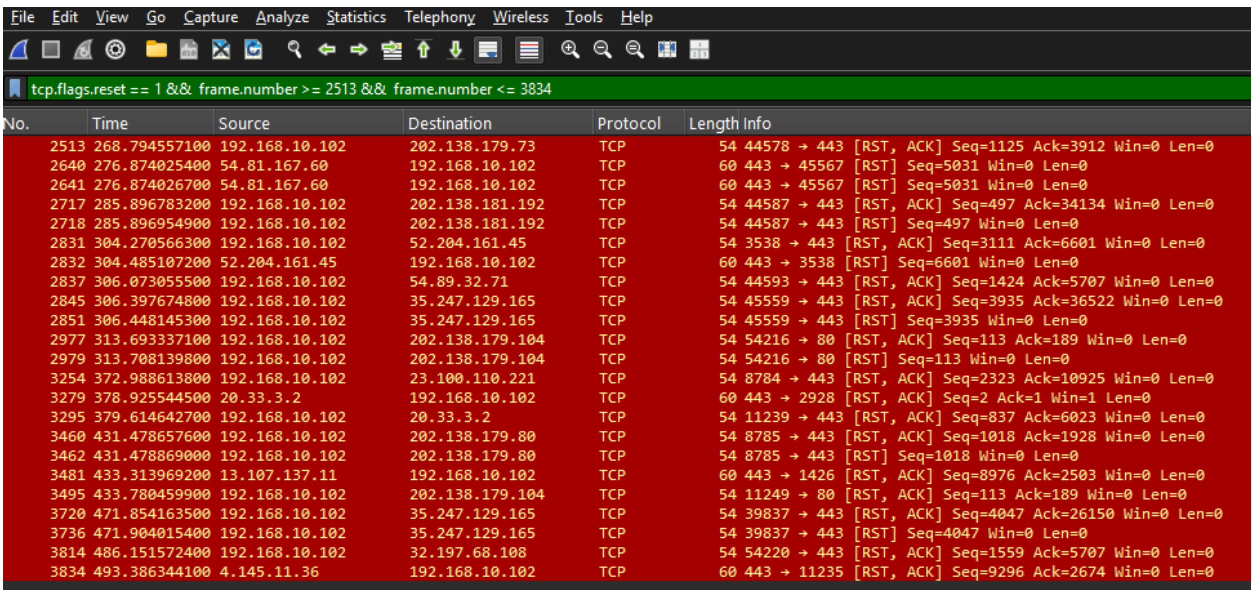
tcp.analysis.duplicate_ack && frame.number >= 1623 && frame.number <= 2512

No.	Time	Source	Destination	Protocol	Length	Info
1671	208.076760100	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1520#1] 11857 → 443 [ACK] Seq=3958 Ack=138 Win=65280 Len=0 SLE=5072 SRE=7912
1674	208.076945900	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1520#2] 11857 → 443 [ACK] Seq=3958 Ack=138 Win=65280 Len=0 SLE=5072 SRE=8982
1675	208.077022400	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1520#3] 11857 → 443 [ACK] Seq=3958 Ack=138 Win=65280 Len=0 SLE=5072 SRE=9549
1678	208.077267400	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1520#4] 11857 → 443 [ACK] Seq=3958 Ack=138 Win=65280 Len=0 SLE=5072 SRE=12389
1687	208.131456000	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1683#1] 11857 → 443 [ACK] Seq=3958 Ack=12389 Win=65280 Len=0 SLE=138 SRE=2978
1690	208.131553900	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1683#2] 11857 → 443 [ACK] Seq=3958 Ack=12389 Win=65280 Len=0 SLE=2978 SRE=4398
1691	208.131592700	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1683#3] 11857 → 443 [ACK] Seq=3958 Ack=12389 Win=65280 Len=0 SLE=4398 SRE=5072
1713	211.089704100	13.33.151.61	192.168.10.102	TCP	60	[TCP Dup ACK 1712#1] 443 → 36736 [ACK] Seq=41 Ack=3 Win=162 Len=0

TCP Duplicate Acknowledgement (Dup ACK) packets were observed 8 times between packets 1671 and 1713.

5. Returning to Home Page Navigation:

Lastly, from the work list page to the home page, the loading time significantly increased to 6 minutes and 35 seconds. This navigation approximately ranges from packets 2513 to 3834 which is a total of 1322 packets captured.



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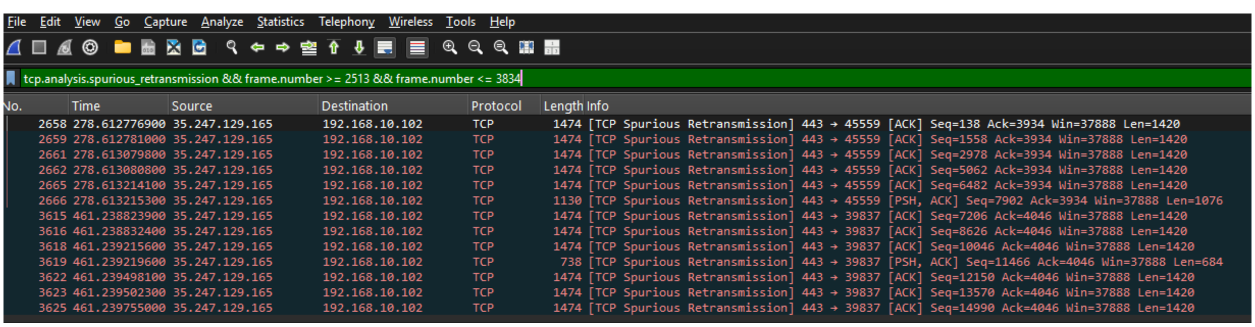
tcp.flags.reset == 1 && frame.number >= 2513 && frame.number <= 3834

No.	Time	Source	Destination	Protocol	Length	Info
2513	268.794557100	192.168.10.102	202.138.179.73	TCP	54	44578 → 443 [RST, ACK] Seq=1125 Ack=3912 Win=0 Len=0
2640	276.874025400	54.81.167.60	192.168.10.102	TCP	60	443 → 45567 [RST] Seq=5031 Win=0 Len=0
2641	276.874026700	54.81.167.60	192.168.10.102	TCP	60	443 → 45567 [RST] Seq=5031 Win=0 Len=0
2717	285.896783200	192.168.10.102	202.138.181.192	TCP	54	44587 → 443 [RST, ACK] Seq=497 Ack=34134 Win=0 Len=0
2718	285.896954900	192.168.10.102	202.138.181.192	TCP	54	44587 → 443 [RST] Seq=497 Win=0 Len=0
2831	304.270566300	192.168.10.102	52.204.161.45	TCP	54	3538 → 443 [RST, ACK] Seq=3111 Ack=6601 Win=0 Len=0
2832	304.485107200	52.204.161.45	192.168.10.102	TCP	60	443 → 3538 [RST] Seq=6601 Win=0 Len=0
2837	306.073055500	192.168.10.102	54.89.32.71	TCP	54	44593 → 443 [RST, ACK] Seq=1424 Ack=5707 Win=0 Len=0
2845	306.397674800	192.168.10.102	35.247.129.165	TCP	54	45559 → 443 [RST, ACK] Seq=3935 Ack=36522 Win=0 Len=0
2851	306.448145300	192.168.10.102	35.247.129.165	TCP	54	45559 → 443 [RST] Seq=3935 Win=0 Len=0
2977	313.693337100	192.168.10.102	202.138.179.104	TCP	54	54216 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
2979	313.708139800	192.168.10.102	202.138.179.104	TCP	54	54216 → 80 [RST] Seq=113 Win=0 Len=0
3254	372.988613800	192.168.10.102	23.100.110.221	TCP	54	8784 → 443 [RST, ACK] Seq=2323 Ack=10925 Win=0 Len=0
3279	378.925544500	20.33.3.2	192.168.10.102	TCP	60	443 → 2928 [RST, ACK] Seq=2 Ack=1 Win=1 Len=0
3295	379.614642700	192.168.10.102	20.33.3.2	TCP	54	11239 → 443 [RST, ACK] Seq=837 Ack=6023 Win=0 Len=0
3460	431.478657600	192.168.10.102	202.138.179.80	TCP	54	8785 → 443 [RST, ACK] Seq=1018 Ack=1928 Win=0 Len=0
3462	431.478869000	192.168.10.102	202.138.179.80	TCP	54	8785 → 443 [RST] Seq=1018 Win=0 Len=0
3481	433.313969200	13.107.137.11	192.168.10.102	TCP	60	443 → 1426 [RST, ACK] Seq=8976 Ack=2503 Win=0 Len=0
3495	433.780459900	192.168.10.102	202.138.179.104	TCP	54	11249 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
3720	471.854163500	192.168.10.102	35.247.129.165	TCP	54	39837 → 443 [RST, ACK] Seq=4047 Ack=26150 Win=0 Len=0
3736	471.904015400	192.168.10.102	35.247.129.165	TCP	54	39837 → 443 [RST] Seq=4047 Win=0 Len=0
3814	486.151572400	192.168.10.102	32.197.68.108	TCP	54	54220 → 443 [RST, ACK] Seq=1559 Ack=5707 Win=0 Len=0
3834	493.386344100	4.145.11.36	192.168.10.102	TCP	60	443 → 11235 [RST, ACK] Seq=9296 Ack=2674 Win=0 Len=0

TCP Reset (RST) packets were observed 23 times between packets 2513 and 3834.



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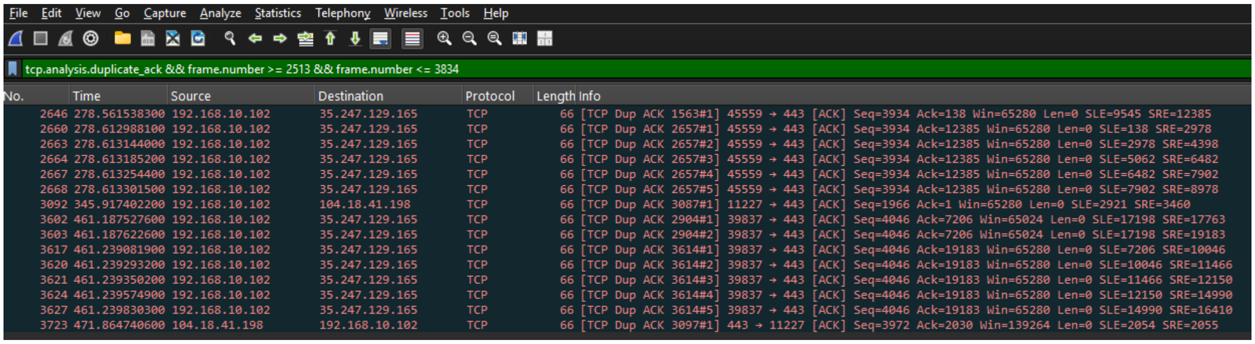
tcp.analysis.spurious_retransmission && frame.number >= 2513 && frame.number <= 3834

No.	Time	Source	Destination	Protocol	Length	Info
2658	278.612776900	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 45559 [ACK] Seq=138 Ack=3934 Win=37888 Len=1420
2659	278.612781000	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 45559 [ACK] Seq=1558 Ack=3934 Win=37888 Len=1420
2661	278.613079800	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 45559 [ACK] Seq=2978 Ack=3934 Win=37888 Len=1420
2662	278.613080800	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 45559 [ACK] Seq=5062 Ack=3934 Win=37888 Len=1420
2665	278.613214100	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 45559 [ACK] Seq=6482 Ack=3934 Win=37888 Len=1420
2666	278.613215300	35.247.129.165	192.168.10.102	TCP	1130	[TCP Spurious Retransmission] 443 → 45559 [PSH, ACK] Seq=7902 Ack=3934 Win=37888 Len=1076
3615	461.238823900	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 39837 [ACK] Seq=7206 Ack=4046 Win=37888 Len=1420
3616	461.238832400	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 39837 [ACK] Seq=8626 Ack=4046 Win=37888 Len=1420
3618	461.239215600	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 39837 [ACK] Seq=10046 Ack=4046 Win=37888 Len=1420
3619	461.239219600	35.247.129.165	192.168.10.102	TCP	738	[TCP Spurious Retransmission] 443 → 39837 [PSH, ACK] Seq=11466 Ack=4046 Win=37888 Len=684
3622	461.239498100	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 39837 [ACK] Seq=12150 Ack=4046 Win=37888 Len=1420
3623	461.239502300	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 39837 [ACK] Seq=13570 Ack=4046 Win=37888 Len=1420
3625	461.239755000	35.247.129.165	192.168.10.102	TCP	1474	[TCP Spurious Retransmission] 443 → 39837 [ACK] Seq=14990 Ack=4046 Win=37888 Len=1420

Spurious Retransmission packets were observed 13 times between packets 2658 and 3625.



No.	Time	Source	Destination	Protocol	Length	Info
2646	278.561538300	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 1563#1] 45559 → 443 [ACK] Seq=3934 Ack=138 Win=65280 Len=0 SLE=9545 SRE=12385
2660	278.612988100	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 2657#1] 45559 → 443 [ACK] Seq=3934 Ack=12385 Win=65280 Len=0 SLE=138 SRE=2978
2663	278.613144000	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 2657#2] 45559 → 443 [ACK] Seq=3934 Ack=12385 Win=65280 Len=0 SLE=2978 SRE=4398
2664	278.613185200	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 2657#3] 45559 → 443 [ACK] Seq=3934 Ack=12385 Win=65280 Len=0 SLE=5062 SRE=6482
2667	278.613254400	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 2657#4] 45559 → 443 [ACK] Seq=3934 Ack=12385 Win=65280 Len=0 SLE=6482 SRE=7902
2668	278.613301500	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 2657#5] 45559 → 443 [ACK] Seq=3934 Ack=12385 Win=65280 Len=0 SLE=7902 SRE=8978
3092	345.917402200	192.168.10.102	104.18.41.198	TCP	66	[TCP Dup ACK 3087#1] 11227 → 443 [ACK] Seq=1966 Ack=1 Win=65280 Len=0 SLE=2921 SRE=3460
3602	461.187527600	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 2904#1] 39837 → 443 [ACK] Seq=4046 Ack=7206 Win=65024 Len=0 SLE=17198 SRE=17763
3603	461.187622600	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 2904#2] 39837 → 443 [ACK] Seq=4046 Ack=7206 Win=65024 Len=0 SLE=17198 SRE=19183
3617	461.239081900	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 3614#1] 39837 → 443 [ACK] Seq=4046 Ack=19183 Win=65280 Len=0 SLE=7206 SRE=10046
3620	461.239293200	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 3614#2] 39837 → 443 [ACK] Seq=4046 Ack=19183 Win=65280 Len=0 SLE=10046 SRE=11466
3621	461.239350200	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 3614#3] 39837 → 443 [ACK] Seq=4046 Ack=19183 Win=65280 Len=0 SLE=11466 SRE=12150
3624	461.239574900	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 3614#4] 39837 → 443 [ACK] Seq=4046 Ack=19183 Win=65280 Len=0 SLE=12150 SRE=14990
3627	461.239830300	192.168.10.102	35.247.129.165	TCP	66	[TCP Dup ACK 3614#5] 39837 → 443 [ACK] Seq=4046 Ack=19183 Win=65280 Len=0 SLE=14990 SRE=16410
3723	471.864740600	104.18.41.198	192.168.10.102	TCP	66	[TCP Dup ACK 3097#1] 443 → 11227 [ACK] Seq=3972 Ack=2030 Win=139264 Len=0 SLE=2054 SRE=2055

TCP Duplicate Acknowledgement (Dup ACK) packets were observed 15 times between packets 2646 and 3723.

Description of Notable Packets

TCP Reset (RST)

TCP Reset (RST) packets indicate that an established communication session between the client and server was unexpectedly terminated. Frequent RST packets may suggest connection interruptions, session terminations, or application-level connection resets.

Spurious Retransmission

Spurious retransmission occurs when a sender retransmits a packet that has already been successfully received and acknowledged by the receiving host. This behavior may indicate network latency, delayed acknowledgements, packet reordering, or inefficient retransmission handling.

TCP Duplicate Acknowledgement (Dup ACK)

Duplicate Acknowledgement packets occur when the receiving host repeatedly acknowledges the same packet sequence number. This behavior commonly indicates packet loss, packet reordering, or delayed packet delivery within the communication path.

Summary

The packet warnings observed throughout the analysis were consistently present across all stages of webpage navigation. TCP Reset (RST), Spurious Retransmission, and TCP Duplicate Acknowledgement (Dup ACK) packets were repeatedly recorded during the captured session. These packet behaviors became more noticeable during stages with prolonged loading times, particularly when navigating back to the Home Page.

Delimitation of Coverage

Due to privacy, confidentiality, and authorized-access restrictions associated with the observed web application, the investigation was limited to client-side packet capture and network traffic observation. Further packet analysis and collection of application-specific data were not conducted. As a result, only a summary of observations is provided in this report. Conclusions, root-cause determinations, and analytical insights were intentionally excluded, as the available information was insufficient to support definitive findings without access to server-side logs, application diagnostics, and other internal system data.